Elicit vs. Teach, Mechanistically: Results on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

Two identical TinyStories-1B twins (same architecture, same pretraining on children's stories, zero arithmetic) are fine-tuned on the same target task — signed 1–4-digit addition/subtraction in natural language, `What is the difference between 8 and 5784?` → `-5776` — with one byte-held protocol (LoRA r=512, α=32, lr 3.53×10^{-4}, batch 128, seed 316, eps/k convergence stop, frozen D_algo_bare, $n = 10^2 \dots 10^6$). The arms differ *only* in the parent checkpoint:

- **Teach twin**: the blank base (`evt-ts1b-base`). The capability is absent.
- **Elicit twin**: **TS1B-latent** (`evt-ts1b-op-bridge-mix`) — the blank base plus (i) a symbol-only arithmetic install (`23 + 45 = 68`, 1M examples, full FT) and (ii) an answer-free word↔symbol binding dose (rewriting `What is the sum of 621 and 5068?` ↔ `621 + 5068`, rehearsed 1:1 with already-seen symbol rows). Every installation is answer-free and leakage-controlled (Table 1). The capability is latent.

Construction step / check	value	control
symbol install: EM on `23 + 45 =`	0.67–0.73	blank: 0.00
corpus census: "sum" in 439M words	21×	"equals" 17, "minus" 57
binding dose: NL→op rewrite exact	0.484	blank + dose: 0.000 EM
sum-only dose on *difference* questions	writes "+"	per-word binding
direct NL exact match of the parent	**0.000**	= blank twin

Table 1: The latent parent is behaviorally identical to the blank twin on the target task before fine-tuning. Everything below is about what differs inside and what that difference does.

Endpoints. Five fine-tuned endpoints enter the results (Table 2). The sweep's LoRA teach endpoints stop by the protocol's own convergence rule far below the elicited child — 0.093 at 1M and 0.139 with four times the unique data — exactly where the paper's Figure 2 puts the TinyStories base (1–2 nats per label token at 1M; "stalls . . . at lower accuracy"). Teaching this base needs full fine-tuning: the recipe that installed the symbol engine gives a taught model at **0.771**. The latent parent was fully fine-tuned with the identical recipe as the method control (0.986). Teach-side results are therefore reported on the *matched full-fine-tune pair* wherever the instrument allows, with the LoRA pair alongside; the three instruments that need adapters (gradient logs, adapter snapshots) exist only for the LoRA runs. The 4M file extends the 1M training set with 3M new unique problems disjoint from every evaluation set; that exclusion exhausts six small operand-length cells, so the extension skews toward 3–4-digit operands (task and evaluation unchanged).

Measurement vocabulary. *Circuit* = top-32 of 528 nodes (512 attention heads + 16 MLP outputs) by attribution score (gradient of the clean/corrupt logit-diff × activation delta, 128–256 pairs);

endpoint	parent	method	data	steps (passes)	stop	EM
elicit LoRA	TS1B-latent	LoRA r=512	1M	11,500 (1.5)	converged	**0.981**
teach LoRA 1M	blank	LoRA r=512	1M	25,000 (3.2)	converged	0.093
teach LoRA 4M	blank	LoRA r=512	4M	39,500 (1.3)	converged	0.139
teach FT	blank	full FT	4M	62,500 (2.0)	ceiling	**0.771**
elicit FT	TS1B-latent	full FT	4M	21,500 (0.7)	converged	**0.986**

Table 2: Fine-tuned endpoints (lr 3.53×10^{-4} LoRA / 2×10^{-5} full FT, batch 128, seed 316, eps/k validation-convergence stop; EM = 0-shot exact match on 1,024 held-out problems). Every endpoint scores $\sim$0 at 16 shots (prompt brittleness of this lineage).

overlaps are Jaccard@32 against a chance floor of 0.031 and a per-map split-half ceiling. *Gradient strength* = the pre-clip global gradient norm $\|g_t\|$ logged at every update (runs' `gradstats.jsonl`); cumulative gradient mass $\sum_t \|g_t\|$. *Weight write* = ΔW = scaling $\cdot\, BA$ per LoRA module (exact); total write $\|\Delta W\|_F$, relative write $\|\Delta W\|/\|W\|$, travel $\|\Delta W_t\|_F$ from adapter snapshots.

Reading guide. The results answer four questions in order: does the *behavior* of learning differ (R1); is the resulting *machinery* the same (R2–R4); how does that machinery *come into being* during training (R5–R8); and what does the latent state *mean and permit* before any target training (R9–R11).

A. Behavior: the learning-curve signature

R1 The learning-curve signature flips.

Metric. Prequential EDL per label token (nats) and held-out exact match, as a function of dataset size n, identical protocol, parent varied.

Result. Figure 1, Table 3. Elicit: EDL strictly monotone, 4.53 $\to$ 0.092; EM 0.54 at n=3,162, 0.98 at 1M. Teach: EDL down–up–down with the hump peaking near n=215K; EM 0.093 at 1M. EM-0.5 threshold: $\sim 3\times10^3$ vs $> 10^6$ ($> 300\times$). A third arm (blank + format-only dose) keeps its hump. Under full fine-tuning the same asymmetry holds without a sweep: the latent parent converges in 21,500 steps (0.7 of a pass; test loss 0.015, EM 0.986), the blank twin runs to the two-pass ceiling (62,500 steps; test loss 0.204, EM 0.771) — and with LoRA reaches only 0.139 at 4M.

Elicit vs. teach. Eliciting spends every example connecting an existing capability (diminishing cost from the first rung); teaching must build the capability, producing increasing returns mid-run. The flip is the paper's causal-intervention claim reproduced on one substrate, with a parent whose latency is itemised rather than assumed. The remaining results explain the flip mechanistically.

B. Machinery: is the circuit the same?

R2 The elicited circuit is the parent's installed circuit; the taught circuit matches nothing upstream.

Metric. Jaccard@32 between attribution circuits (chance 0.031; ceiling = split-half self-agreement of the map, 0.684 for the elicited child).

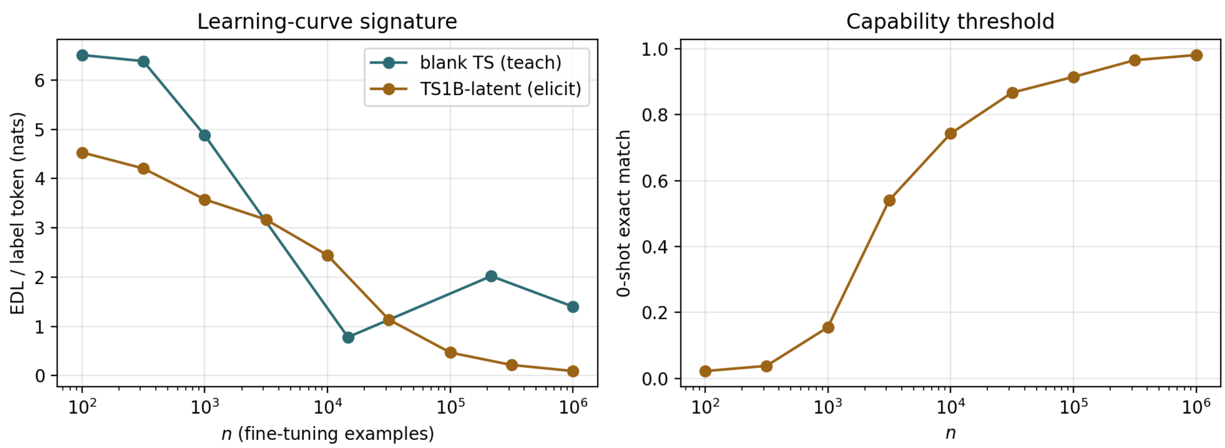

Figure 1: EDL/token (left) and 0-shot EM (right) vs n; teal = teach twin, gold = elicit twin.

n	elicit EDL/tok	elicit EM	teach EDL/tok	teach EM
100	4.533	0.023	6.511	0.007
316	4.208	0.038	6.387	0.033
1,000	3.579	0.155	∼4.9	∼0
3,162	3.168	**0.541**	↓	∼0
10,000	2.444	0.743	min ∼0.8	∼0
31,623	1.137	0.867	↑	∼0
100,000	0.467	0.915	↑	∼0
316,228	0.215	0.966	peak ∼2.0 @215K	∼0
1,000,000	**0.092**	**0.981**	1.40	0.093

Table 3: Per-rung values (EDL nats/label token; EM on 1,024 held-out problems).

Result. Table 4. Installed engine ↔ elicited child: 0.455–0.524 across eval surfaces (≈70% of ceiling), top-4 nodes identical and identically ordered (`mlp:15,14,13,12`); the engine backbone is already fully shared at n=3K (child@3K ↔ child@1M 0.391, with only the input interface churning). Blank parent: no measurable circuit (logit-diff −0.33 even 16-shot; its map is noise), so the taught child has nothing to match. The elicited circuit is method-independent: the fully fine-tuned elicited child matches the LoRA-elicited child at 0.641 (ceilings 0.56 / 0.68).

Elicit vs. teach. Elicitation re-weights machinery that was already there; teaching constructs machinery where there was none. "Reuse" is measured, not inferred.

R3 The parent's functional head roles survive elicitation almost intact; a performing taught model rebuilds most of them; a failing one has none.

Metric. Desiderata Component Masking (Prakash et al. 2024), heads only: for each functional variable — operand a, operand b, operation ($+ \leftrightarrow -$) — learn the sparse set of attention heads whose counterfactual activations, patched into the clean run, flip the answer to the counterfactual one; compare role sets across models by Jaccard (chance ≈ 0.05).

Result. Table 5. Parent (installed engine) vs LoRA-elicited child, cf-flip at the ceiling in every set: operand-a 0.879, operand-b 0.900, operation 0.750; the operand roles are essentially

Comparison	J@32	chance	ceiling
installed engine ↔ elicited child (op / bridge surface)	0.455–0.488 / 0.524	0.031	0.684
elicited child @3K ↔ @1M	0.391	0.031	0.684
blank parent ↔ taught child	undefined (parent map is noise)	–	–
elicited child, full FT ↔ elicited child, LoRA	**0.641** (Spearman 0.54)	0.031	0.561 / 0.684
elicited ↔ taught, full-FT pair	0.333 (Spearman 0.16)	0.031	0.561 / 0.422
elicited ↔ taught, LoRA pair (4M)	0.391 (Spearman 0.28)	0.031	0.684 / 0.524
elicited ↔ taught, LoRA pair (1M)	0.231 (Spearman −0.11)	0.031	0.684 / 0.561
taught, full FT ↔ taught, LoRA 4M	0.391 (Spearman 0.11)	0.031	0.422 / 0.524

Table 4: Circuit-membership overlaps. All maps pass the performing-regime guard.

the whole of layer-0 attention. The fully fine-tuned elicited child keeps the same roles (0.811 / 0.833 vs the LoRA child; 0.694 / 0.857 vs the parent). The fully fine-tuned taught child (EM 0.77) also has compact fetcher sets at the ceiling, overlapping the elicited child's at 0.587 / 0.763 and the parent's at 0.763 / 0.684; the two fully fine-tuned children overlap each other at 0.700 / 0.632. Both LoRA-taught children (EM 0.09 / 0.14) have no compact set: cf-flip 0.008–0.04 against ceilings of 0.14–0.20, overlaps ≤ 0.13.

Elicit vs. teach. Operand reading is a layer-0 function of the substrate, so every model that solves the task uses largely the same fetcher heads. What the regime changes is how completely: elicitation keeps the parent's roles at 0.8–0.9, independent teaching reconverges on them at 0.6–0.76 — a graded difference, and a smaller one than circuit membership. The earlier categorical reading ("taught roles do not overlap; no compact set") was a property of taught models that had not learned the task.

Comparison (operand a / operand b)	$\lvert A\rvert$	$\lvert B\rvert$	Jaccard	cf-flip / ceiling
parent ↔ elicited (LoRA), op surface	30 / 27	32 / 30	**0.879 / 0.900**	at ceiling
parent ↔ elicited (LoRA), operation role	25	24	**0.750**	at ceiling
elicited (full FT) ↔ elicited (LoRA), NL	31 / 25	36 / 30	0.811 / 0.833	0.98 / 0.98
elicited (full FT) ↔ parent	31 / 25	30 / 27	0.694 / 0.857	–
taught (full FT) ↔ elicited (LoRA), NL	37 / 37	36 / 30	0.587 / 0.763	0.81 / 0.79 (ceiling 0.81 / 0.80)
taught (full FT) ↔ parent	37 / 37	30 / 27	0.763 / 0.684	–
taught (full FT) ↔ elicited (full FT), NL	37 / 37	31 / 25	0.700 / 0.632	–
taught (LoRA 4M) ↔ elicited (LoRA), NL	12 / 21	36 / 30	0.043 / 0.041	0.01 / 0.03 (ceiling 0.20 / 0.14)
taught (LoRA 1M) ↔ elicited (LoRA), NL	9 / 21	36 / 30	0.125 / 0.085	0.02 / 0.04 (ceiling 0.15)

Table 5: DCM head roles (heads only, 64–128 counterfactual pairs). "At ceiling" = cf-flip accuracy equals the full-counterfactual ceiling.

R4 Same task, same substrate: elicitation lands on one machine, teaching on a related but different one — and the regime, not the method, decides.

Metric. Layer distribution of the top-16 attribution nodes; J@32 and score Spearman between children, each against the two maps' split-half ceilings; for the LoRA-1M pair also the edge-level change rate $\Delta S = 1 - \text{Jaccard}$ (edges @256, node→block attribution).

Result. Figure 2, Table 4. The two elicited children — one LoRA, one fully fine-tuned — share one circuit (0.641, at the ceiling). Every taught child sits apart from it: 0.333 for the matched full-FT pair, 0.391 for the LoRA-4M pair, 0.231 for the LoRA-1M pair (≈60% of the geometric-mean ceiling), with score correlations −0.11 to 0.28; the two taught children agree with each other at 0.391. What is shared is the late-MLP engine (`mlp:15,14,12/13,7`); what differs is the rest: the taught circuits add `mlp:0,10,11` and their own attention — eight layer-0 heads over raw digit tokens under LoRA, layer-5/6/15 heads under full FT — where the elicited circuits use `mlp:8,9` and the layer-7 trio. LoRA-1M pair: $\Delta S_{\text{node}} = 0.769$, $\Delta S_{\text{edge}} = 0.739$, both far above the noise floors (0.32 / 0.61).

Elicit vs. teach. Because every child shares the identical pretrained substrate, the mechanism difference is caused by the learning regime; and because it survives a change of method on both sides, it is not an artefact of the adapter. It is a partial difference: teaching reconverges on the same late MLPs but fetches and routes operands through machinery of its own, so these are two related machines rather than one machine re-routed or two unrelated ones. The earlier "total" reading came from a taught model that had not learned the task.

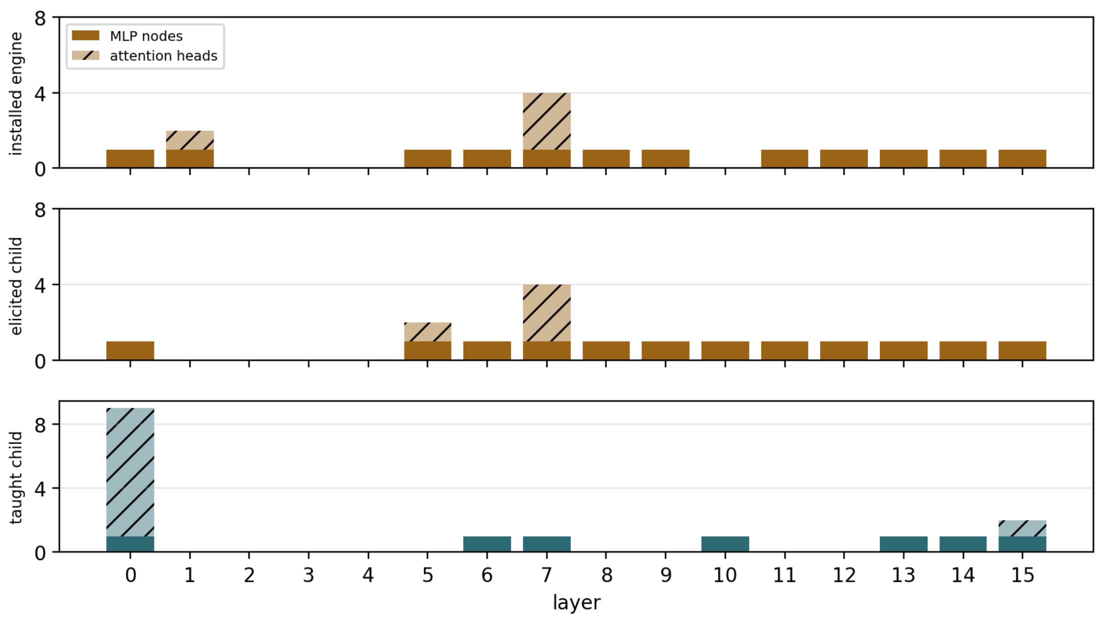

Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the installed engine, the LoRA-elicited child, and the LoRA-1M taught child.

C. Dynamics: how the machinery comes into being

R5 The elicited circuit exists before training; the taught circuit is built mid-run — through the EDL hump.

Metric. Jaccard@32 of each training snapshot's circuit against the run's final circuit.

Result. Figure 3. Elicit: 0.488 at step 1 (16× chance), a brief interface-wiring transient (dip to ∼0.42 with a strong map at step 154), locked from ∼step 1.5K of 11.5K (0.561; 82% of ceiling, 97% of the J@64 ceiling by 8.4K). Teach: noise floor (≈0.22–0.28) until ∼step 1.3K, then 0.27 → 0.44 over steps 1.3K–5.4K — the region where its EDL peaks.

Elicit vs. teach. Elicitation finds and wires; teaching constructs. The elicited circuit finishes stabilising before the taught one begins to exist, and the teaching hump of R1 is literally the circuit being assembled.

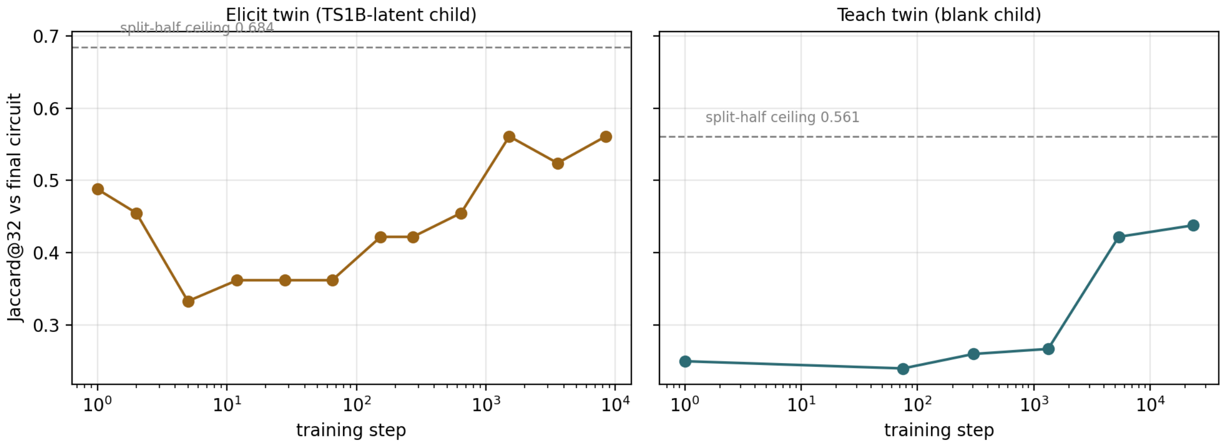

Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

R6 Elicitation's gradient dies; teaching's grows.

Metric. Pre-clip global gradient norm at every update, from the runs' `gradstats` logs: peak, mean over first 1% / middle / last 10% of steps, cumulative gradient mass $\sum_t \|g_t\|$, and its split across weight classes.

Result. Table 6, Figure 4. Both arms start at the same scale (first-1% mean 0.18 vs 0.17). Elicit then decays ×3.3 to a floor of 0.056 (peak 0.44 at step 96). Teach *grows* ∼40× to 6.8 (peak 17.4 at step 23,657 — still rising when the run stopped). Cumulative gradient mass 783 vs 116,877 (∼150×). Across n the arms are indistinguishable up to 10K and diverge from 31.6K on (mass ratio 7× → 55× → 95× → 150×); the 4M teach run continues the trend (peak 22.0 at step 36,931, last-10% mean 7.8, mass 220,685, ∼280×). Teach's gradient mass migrates out of the MLPs (share 0.47 → 0.01) into attention (VO 0.87; 0.93 at 4M).

Elicit vs. teach. The optimizer's view of R5: eliciting converges and its gradient dies; teaching keeps generating gradient pressure that grows with the structure being built, aimed at attention — the layer-0 army of R4 being written. This is the clearest dynamical discriminator, and it sets up R7: what all that gradient actually bought.

LoRA fine-tune	elicit 1M	teach 1M	teach 4M
peak grad norm (step)	0.441 (96)	17.4 (23,657)	22.0 (36,931)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80	0.160 / 6.61 / 7.78
first 1% ÷ last 10%	×3.3 (decays)	×0.02 (grows ∼40×)	×0.02 (grows ∼50×)
cumulative gradient mass $\sum_t \|g_t\|$	783	116,877	220,685
gradient mass per step	0.068	4.68	5.59
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01	.05 / .93 / .01

Table 6: Raw gradient strength of the LoRA fine-tunes (pre-clip global norm at every update).

R7 Teaching writes more into the weights and buys far less per unit written.

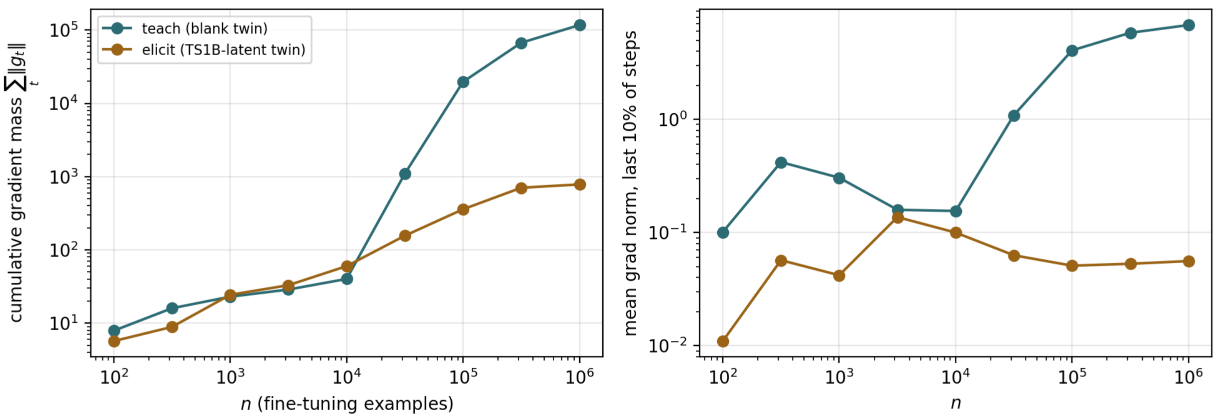

Figure 4: Cumulative gradient mass (left) and late-run gradient norm (right) vs n; teal = teach, gold = elicit.

Metric. Relative write $\|\Delta W\|/\|W\|$ at n=1M (exact from LoRA factors); total travel $\|\Delta W_t\|_F$ from adapter snapshots; capability per unit travel = final EM / total travel.

Result. Table 7. Relative write 0.212 (teach) vs 0.095 (elicit), 2.2×; total travel 457 vs 120 (3.8×) over 2× as many steps (23.5K vs 11.5K); capability per unit travel 2.0×10^{-4} vs 8.1×10^{-3}, ∼40×. The travel *profiles* of the two arms coincide (both burst then decay): AdamW's $\sqrt{v}$-normalisation compresses R6's 150× gradient-mass gap into a 3.8× step-size gap with a shared shape. The matched full-FT pair repeats the magnitude gap without adapters: relative write 0.094 (teach) vs 0.050 (elicit), 1.9×, with identical effective rank (571 vs 554 of 2048).

Elicit vs. teach. What R6's gradients bought: teaching writes 2–4× more into the weights yet buys ∼40× less capability per unit written — eliciting *connects* what exists, teaching *writes* what does not. Update size is the muted echo of the gradient signal; magnitude and efficiency, not timing, are where the weights show the regime.

	elicit (TS1B-latent)	teach (blank)
relative write $\|\Delta W\|/\|W\|$, LoRA 1M	0.095	0.212
relative write, LoRA 4M teach / full-FT pair (4M)	0.050 (FT)	0.264 / 0.094 (FT)
steps to convergence	11,500	23,496
peak / final writing speed (per step)	0.218 / 0.018	0.306 / 0.022
total weight travel $\|\Delta W\|_F$	120	457
final exact match	0.981	0.093
capability per unit travel	8.1×10^{-3}	2.0×10^{-4}

Table 7: Weight-space write, elicit vs teach (travel curves in `05_weights/fig_weight_travel`).

R8 Elicitation's shift is per-problem content; teaching's shift is half a shared direction.

Metric. Residual-stream shift between parent and child on identical inputs (both models in fp32): $\|h_\ell^{\text{child}} - h_\ell^{\text{parent}}\|/\|h_\ell^{\text{parent}}\|$ per layer at the answer position of 256 NL problems and on 256 held-out TinyStories passages; direction statistics of the 256 per-problem shift vectors at the read-out — energy fraction of the top singular direction (PC1) and cosine to the mean — next to the same statistic on the parent's own states (the anisotropy reference), and again after projecting out the unembedding directions of the answer token; functional shift = KL(parent ‖ child) at the answer position vs per token on stories, and the change in story loss.

Result. Figure 5, Table 8. Direction at the read-out is method-independent: elicit PC1 0.063 (LoRA) / 0.056 (full FT) with the parent's own states at 0.23 — no dominant shared direction; teach PC1 0.528 (LoRA 4M) / 0.521 (full FT), and still 0.52 after the answer-token directions are removed. Cosine-to-mean: 0.26 / 0.23 vs 0.26 / 0.74. Magnitude is method-dependent: the LoRA-taught final-layer write is 14–29× the parent's norm, the fully fine-tuned one 1.4× (elicit 1.7–1.9×). Generic text: within LoRA the taught child displaces stories 5–9× more (KL 0.33 vs 0.07 nats/token; story loss +0.32 vs +0.04); within full FT both displace equally (+1.13 vs +1.04), the method dominating.

Elicit vs. teach. The robust regime signature is what the shift carries. The elicited child differs from its parent by a per-problem vector — the answer routed to the output — because the parent already sits in a problem-specific state (R9); the taught child's shift is dominated by one shared component because the blank parent's answer-position state is problem-blind and the mode itself has to be moved before any content can. That is why the mean-vector patch in R11 yields format only on the elicit side. Norm-based "loud write" and generic-displacement readings depend on the method and are not regime signatures.

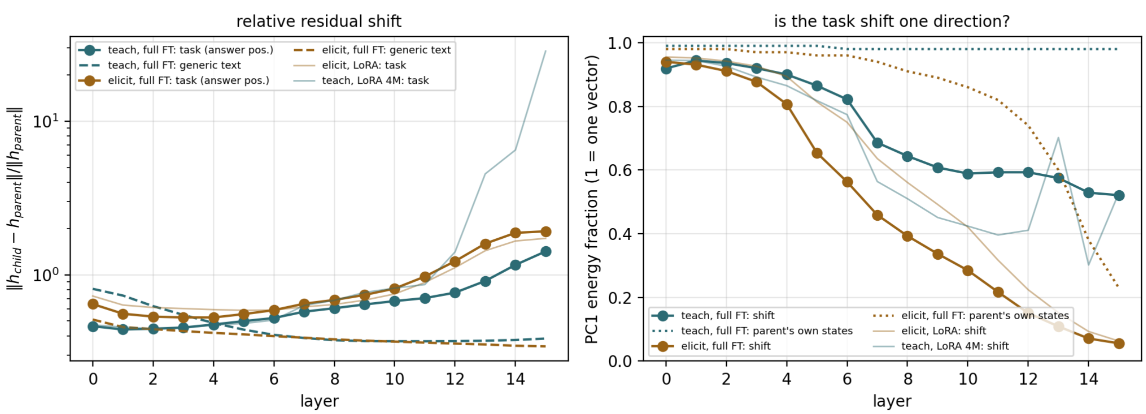

Figure 5: Matched full-FT pair. Left: relative residual shift by layer at the answer position (solid) and on generic text (dashed). Right: PC1 energy fraction of the per-problem shift vectors, with the parent-state reference dotted.

D. The latent state: what it means and what it permits before training

R9 In the latent parent, the words already reach the engine.

Metric. Cross-format activation matching: per node, mean cosine between the activation on the NL rendering and on the symbol rendering of the *same* problem, minus the mismatched-pair baseline ("does `What is the sum of 621 and 5068?` drive the node into the same problem-specific state as `621 + 5068 = ?`").

	elicit LoRA 1M	teach LoRA 4M	elicit FT	teach FT
final exact match	0.981	0.139	0.986	0.771
shift at answer position, final layer	1.72	28.6	1.92	1.42
shift on generic text, layer mean	0.160	0.236	0.396	0.464
KL(parent‖child) at answer position (nats)	14.3	14.0	16.4	22.2
KL(parent‖child) on stories (nats/token)	0.067	0.326	0.721	1.147
story loss, child − parent (nats/token)	+0.035	+0.315	+1.038	+1.133
PC1 of shift, final layer (parent states)	**0.063** (0.23)	**0.528** (0.98)	**0.056** (0.23)	**0.521** (0.98)
PC1 after removing answer-token directions	0.072	0.520	0.061	0.520
cosine-to-mean, final layer	0.26	0.26	0.23	0.74

Table 8: Residual-stream shift parent→child (N=256; generic text = 256 held-out TinyStories passages, 64 tokens). LoRA-1M teach: final-layer shift 14.1, PC1 0.351, story loss +0.15; construction (blank→latent, full FT): generic shift 1.12, PC1 0.60.

Result. Engine without the binding dose: MLP index ≈ 0.00–0.01 at every layer. TS1B-latent (binding installed): +0.07–0.13 across layers 7–15, ∼10×. Attention-head indices are confounded by shared digit tokens and are excluded. The geometry of the answer-position state agrees (final layer, 256 problems): the latent parent's states have cosine-to-mean 0.51 and PC1 0.23 — problem-specific — while the blank base's have 0.99 and 0.98 — problem-blind.

Elicit vs. teach. This is what "latent" means internally: before any target training, NL input already activates the arithmetic MLPs of the elicit parent, while the teach parent has no arithmetic MLPs to activate. It also confirms the binding dose installed an access route, not the task.

R10 The latent parent already holds the answer in its verbalizable space; the blank base holds nothing. Elicitation forms the answer where the engine forms it; teaching forms it only at the last layer.

Metric. Logit lens, Jacobian lens (J-lens: the residual at layer ℓ transported to the final-layer basis by the corpus-average Jacobian $J_\ell = \mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$ over held-out stories, then unembedded) and R-lens (the J-lens with a relevance-propagation backward pass), at the answer position of 256 problems. Scored token = first digit chunk of the answer (sign appended to the prompt for negative answers). Read-outs: median rank of the answer token (chance 64K of 128K), logit-diff against a mismatched problem's answer, settled depth (first layer from which the token stays top-1).

Result. Figure 6, Table 9. Latent parent on the NL target, J-space rank of the answer: 2,606 at layer 12, 510 at layer 14, 174 in the output distribution — while it emits a rewrite; logit-diff rising from layer 8. Blank base: 55–80K at every layer, logit-diff 0.00 everywhere, in all three lenses. Elicited child: rank 139 / 6 / 1 at layers 12 / 13 / 14, settled median layer 14 — the same trajectory as the parent's engine on symbols (122 / 6 / 1) and the same onset as the parent on NL (layer 8). LoRA-taught children: rank 41,677 / 19,933 / 8 (1M) and 41,722 / 17,183 / 6 (4M) at layers 12 / 14 / 15. The matched full-FT pair repeats it with two performing models: elicit rank 338 / 8 / 1 at layers 12 / 13 / 14, settled at layer 14 (quartiles 14–14), top-1 0.80 at layer 14; teach rank 4,196 / 1,916 / 79, settled at layer 15 (quartiles 15–15), top-1 0.12 at layer 14 → 0.80 at layer 15. Onset (logit-diff > 1) is layer 8–9 for both.

Elicit vs. teach. "Latent" in the workspace sense: the answer is present and rising in the parent's verbalizable space before any target training, and elicitation amplifies that late read-out without moving it — parent and child lens trajectories coincide layer by layer on the engine's own surface. Teaching has no signal to amplify: its answer exists nowhere in the stack until the last layer writes it — and a taught model that does solve the task still computes the answer one layer later than the engine, under either method. Elicit vs. teach: depth is inherited from the engine under elicitation and pinned to the last layer under teaching. The "middle layers" form of the prediction does not hold — formation is at layers 13–14 of 16 in parent and elicited child alike.

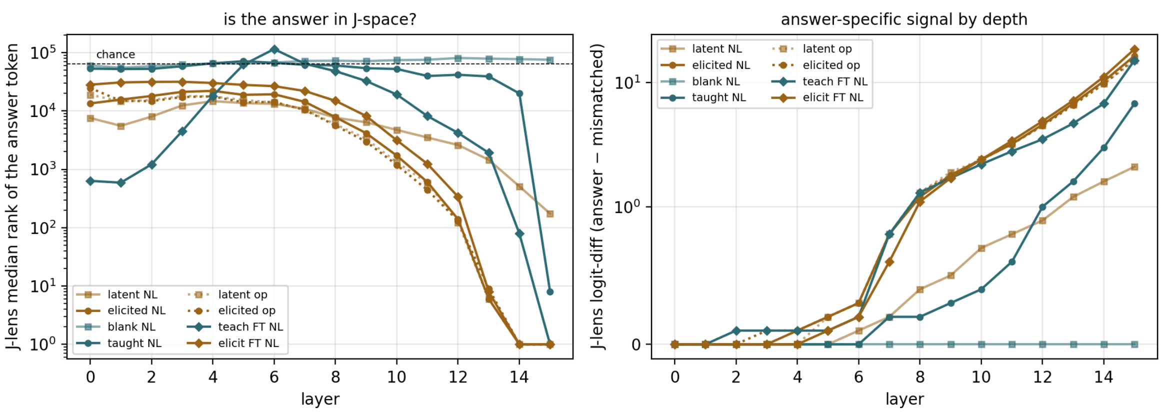

Figure 6: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff (right). Squares = parents, circles = LoRA children, diamonds = the full-FT pair; dotted = symbol surface.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out
TS1B-latent (parent)	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174
elicited child	NL	0.957	1,584 / 139 / 464	1 / 1 / 1	1
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441
taught child (LoRA 1M)	NL	0.109	51,167 / 41,677 / 44,335	35,620 / 19,933 / 21,156	8
taught child (LoRA 4M)	NL	0.117	54,896 / 41,722 / 41,666	40,496 / 17,183 / 21,998	6
taught child (full FT)	NL	0.797	18,502 / 4,196 / 9,429	141 / 79 / 377	1
elicited child (full FT)	NL	0.988	1,797 / 338 / 492	1 / 1 / 1	1
TS1B-latent (parent)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1
elicited child	symbols	0.855	1,360 / 131 / 399	1 / 1 / 1	1

Table 9: Lens depth of the answer's first digit token: median rank among 128K entries (chance 64K) under the logit lens / J-lens / R-lens; "own acc" = the model's own first-digit-token accuracy on the 256 held-out problems; read-out = last layer, where all lenses coincide with the model's output.

R11 The latent capability can be reached with frozen weights; the absent one cannot.

Metric. Exact match on the target task under zero-training interventions: (a) self-chain — the model rewrites the question itself, then computes its own rewrite; (b) donor patching — the fine-tuned child's activations at the 32 circuit nodes written into the parent (mean vector, or per-prompt states); random-node controls.

Result. Figure 7, Table 10. Elicit parent: direct 0.000 → self-chain **0.328** (= 0.484 × 0.668; the composition law parse×compute held on four model variants); per-prompt states 0.125 (random nodes: 0.000); mean vector: format only. Teach side: blank + same doses 0.000; blank + chain 0.000 (a parser without an engine); taught donor into blank twin 0.000 in every condition — for the LoRA-1M, LoRA-4M and full-FT taught donors alike. The full-FT elicited donor reproduces the elicit side: per-prompt states 0.121 (format 0.387), mean vector format only.

Elicit vs. teach. Elicit = one composition step or one state-patch away from the target; teach = nothing underneath to switch on. The teach-side nulls double as the leakage control for the whole construction — and they close the loop with R2: the machinery the patch activates is the same installed engine the fine-tune later reuses.

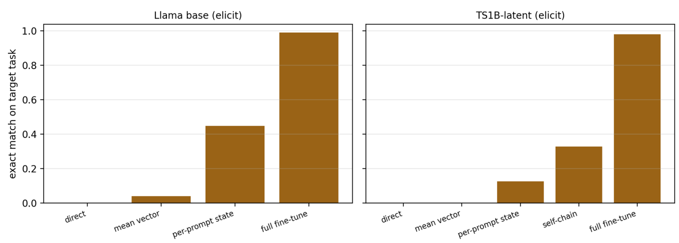

Figure 7: Intervention ladder on the elicit parent (weights frozen except the last bar).

Patient + intervention (k=32 circuit nodes)	condition	format	EM
TS1B-latent + child mean shift	circuit nodes	0.164	0.000
TS1B-latent + child per-prompt states	circuit nodes	0.359	**0.125**
	random-32 nodes	0.051	0.000
TS1B-latent self-chain (no donor)	own rewrite + "= "	–	**0.328**
blank + binding dose, self-chain	parser without engine	–	0.000
TS1B-latent + full-FT elicited donor, per-prompt states	circuit nodes	0.387	**0.121**
blank + taught donor (LoRA 1M / LoRA 4M / full FT)	every condition	$\leq$0.02	0.000

Table 10: Zero-training interventions, TinyStories twins.

E. The matched full-fine-tune pair at a glance

Both regimes under one method: the blank twin and the latent twin, each fully fine-tuned with the identical recipe on the identical 4M file. The one difference is the parent. Every row is a result above, restricted to this pair.

Validation results (not regime differences, but what licenses R2–R11)

- **Every circuit is real and load-bearing.** True activation patching, top-8 of 528 nodes sufficient / necessary: LoRA-elicited 0.997 / 0.998, LoRA-1M taught – / 0.986, LoRA-4M taught 0.976 / 0.987, full-FT taught 0.950 / 0.966, full-FT elicited 0.923 / 0.943 (top-32 $\geq$ 0.997 everywhere). Every overlap above compares provably load-bearing sets.

	elicit FT	teach FT	result
exact match (0-shot)	0.986	0.771	R1
steps to stop (passes)	21,500 (0.7), converged	62,500 (2.0), ceiling	R1
held-out test loss (nats/token)	0.015	0.204	R1
circuit J@32 vs the other member (ceilings 0.56 / 0.42)	0.333, Spearman 0.16		R4
circuit J@32 vs the LoRA-elicited child	0.641	0.306	R2, R4
top-8 nodes sufficient / necessary	0.923 / 0.943	0.950 / 0.966	validati
DCM operand roles vs the parent's engine (J, a / b)	0.694 / 0.857	0.763 / 0.684	R3
DCM operand roles vs the other member (J, a / b)	0.700 / 0.632		R3
relative weight write $\|\Delta W\|/\|W\|$	0.050	0.094	R7
effective rank of ΔW (of 2048)	554	571	non-dis
final-layer residual shift at the answer position	1.92	1.42	R8
PC1 of the shift at the read-out (after removing answer content)	0.056 (0.061)	0.521 (0.520)	R8
story loss, child − parent (nats/token)	+1.04	+1.13	non-dis
answer settled depth, median (quartiles)	L14 (14–14)	L15 (15–15)	R10
J-space rank of the answer at layers 12 / 13 / 14	338 / 8 / 1	4,196 / 1,916 / 79	R10
donor states into own parent, per-prompt (mean vector)	0.121 (format only)	0.000 (0.000)	R11

Table 11: The matched full-FT pair. The rows that discriminate the regimes under one method: exact match and convergence, circuit identity, write magnitude, shift direction, settled depth, donor patching. Effective rank, story-loss displacement and DCM overlap are listed for completeness and do not.

- **Teach-side replication on performing endpoints.** Every teach-side measurement was re-run on the LoRA-4M and full-FT taught endpoints, with the full-FT elicited endpoint as the method control. Three earlier readings did not survive and are reworded above: taught roles "absent" (R3), the machine difference "total" (R4), and the taught write "loud" (R8). The rest replicate: signature (R1), engine reuse (R2), gradient growth (R6), write magnitude (R7), per-problem vs shared shift direction (R8), last-layer formation (R10), donor nulls (R11).
- **Lens depth is scored on the first digit token.** Scoring the raw first token credited the taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign appended to the prompt the effect vanishes (first-token accuracy 0.31 → 0.11; no early route); the other three models were unaffected.
- **Construction controls.** Identical doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves "difference" unbound (per-word binding); the composition law chain = parse × compute held on four variants.